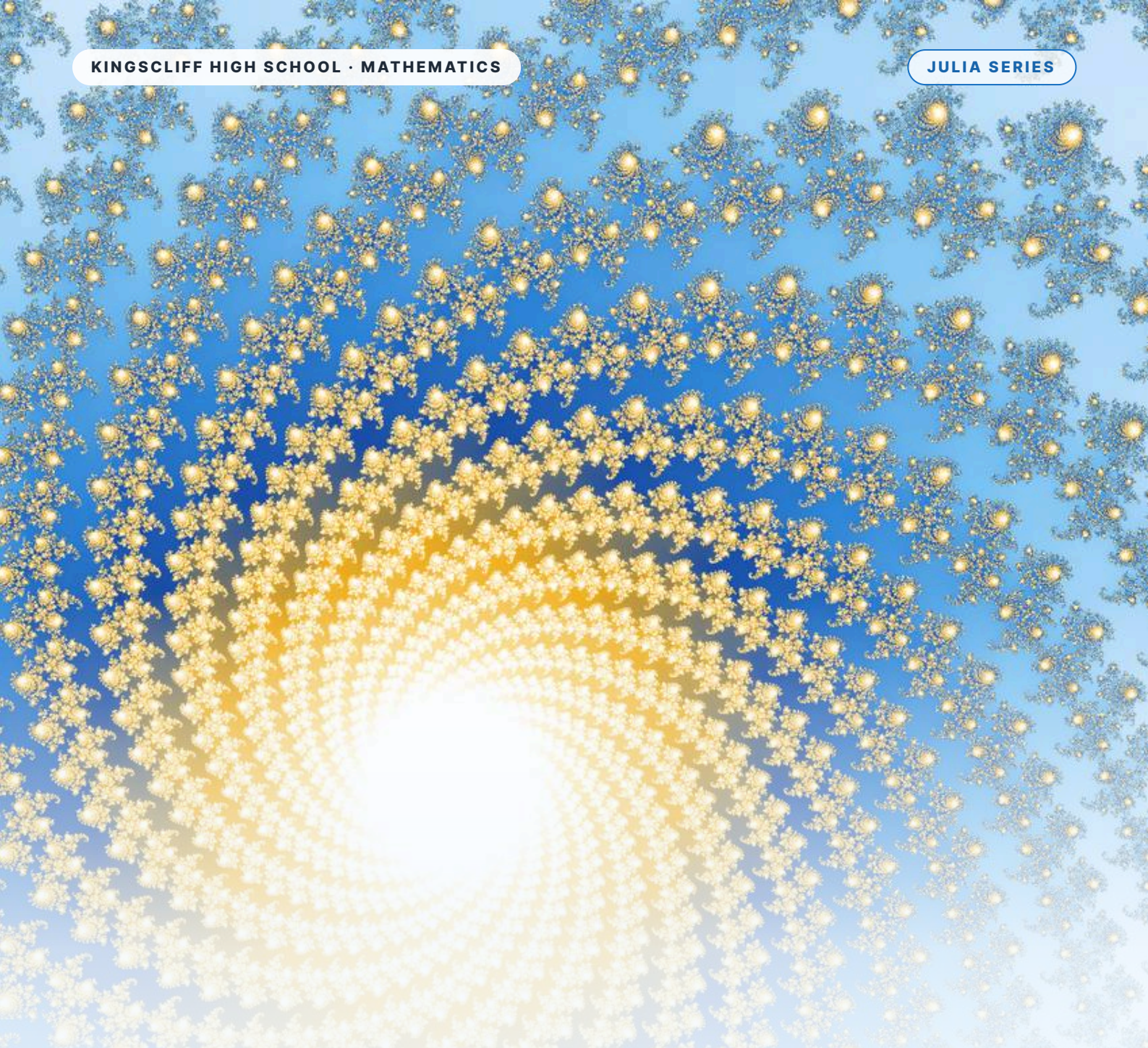


KINGSCLIFF HIGH SCHOOL · MATHEMATICS

JULIA SERIES

YEAR 9 ADVANCED → YEAR 10 ADVANCED

Summer Maths

Ready for Year 10

Six weeks. Twenty questions a week. Every Year 9 skill that Year 10 Advanced builds on – mixed together so it sticks.

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6

NAME _____ CLASS _____

How it works

One page a week

Each week has **20 questions** and a **bonus**. Set aside about **40 minutes** – one sitting or a few short ones. Questions start easier and get a little harder through the page.

The same pattern every week

Question 1 is always number and percentages, question 11 is always Pythagoras, and so on. If the same number keeps going wrong, that is the skill to revise before Year 10.

1	Number & percentages	11	Pythagoras
2	Scientific notation	12	Trigonometry
3	Indices	13	Surface area & volume
4	Surds	14	Distance & midpoint
5	Algebraic fractions	15	Gradient
6	Expand	16	Equation of a line
7	Factorise	17	Geometry
8	Linear equations	18	Probability
9	Quadratic equations	19	Financial maths
10	Formulas & inequalities	20	Problem solving

Why Julia? Each banner is part of the same *Julia set*: pick a number c, then repeat $z \to z^2 + c$ for every point. Points that stay small are coloured in. Every booklet in the series zooms into a different corner of the same set – this one dives into the heart of one of its two big spirals.

NSW syllabus (NESA Mathematics K–10, 2022). Revises MA5-ALG-C-01, MA5-ALG-P-01, MA5-ALG-P-02, MA5-IND-C-01, MA5-IND-P-01, MA5-IND-P-02, MA5-MAG-C-01, MA5-EQU-C-01, MA5-EQU-P-01, MA5-EQU-P-02, MA5-LIN-C-01, MA5-LIN-C-02, MA5-TRG-C-01, MA5-TRG-C-02, MA5-ARE-C-01, MA5-VOL-C-01, MA5-GEO-C-01, MA5-GEO-P-01, MA5-PRO-C-01, MA5-PRO-P-01, MA5-FIN-C-01, with MA4-PYT-C-01. ★ Year 10 preview: MA5-FIN-C-02, MA5-LIN-P-01, MA5-TRG-P-01, MA5-TRG-P-02, MA5-VOL-P-01, MA5-POL-P-01, MA5-LOG-P-01, MA5-CIR-P-01.

Rules of the game

- **No calculator.** Leave answers exact (e.g. $3\sqrt{5}$ or 96π) unless asked to round. Trig questions give you any value you need.
- **Show working** in the space under each question.
- **Mark it yourself** using the answers at the back (pages 10–12). Each answer has a line of working to help you find a mistake.
- **Fix it.** Redo anything you got wrong a day or two later, without looking.
- ★ questions preview Year 10 Advanced work. Have a go – they are the ones that will help most in Term 1.
- **Stuck?** The hints page (page 9) has the key facts and formulas.

My summer tracker

WEEK	DATE	SCORE	DONE	BONUS	REVISE (Q NUMBERS)
1 Week 1		/ 20	☐	☐	
2 Week 2		/ 20	☐	☐	
3 Week 3		/ 20	☐	☐	
4 Week 4		/ 20	☐	☐	
5 Week 5		/ 20	☐	☐	
6 Week 6		/ 20	☐	☐	

First up in Year 10 Advanced: interest and depreciation, graphing lines, surface area and volume, products, factors and surds, and equations, inequalities and logarithms. Every one of them starts from the skills in these six pages.

WEEK 1

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & surds · Algebra · Equations · Measurement & trig · Coordinate geometry · Geometry & chance · In context · ★ Year 10 preview (Stage 5 path) · No calculator

1 NUMBER & PERCENTAGES

Decrease \$360 by 15%.

2 SCIENTIFIC NOTATION

Write $0.000\,058$ in scientific notation.

3 INDICES

Simplify $(3x^2y)^3$

4 SURDS

Simplify $\sqrt{48}$

5 ALGEBRAIC FRACTIONS

Simplify $\frac{x}{3} + \frac{x}{4}$

6 EXPAND

Expand $(x+7)(x-3)$

7 FACTORISE

Factorise $x^2 + 9x + 20$

8 LINEAR EQUATIONS

Solve $\frac{x+3}{4} = x - 3$

9 QUADRATIC EQUATIONS

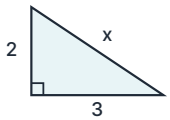
Solve $x^2 - 7x + 10 = 0$

10 FORMULAS & INEQUALITIES

Make h the subject of $V = \pi r^2 h$.

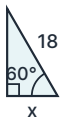
11 PYTHAGORAS

Find x as an exact value.



12 TRIGONOMETRY

Use $\cos 60^\circ = \frac{1}{2}$ to find x. (cm)



13 SURFACE AREA & VOLUME

Find the volume of a cylinder with radius 5 cm and height 4 cm, in terms of π.

14 DISTANCE & MIDPOINT

Find the distance between $(-1,\ 3)$ and $(5,\ 11)$.

15 GRADIENT

Find the gradient of the line through $(-2,\ 7)$ and $(4,\ -5)$.

16 EQUATION OF A LINE

Find the equation of the line with gradient 2 that passes through $(1,\ 5)$.

17 GEOMETRY

Find the size of each interior angle of a regular hexagon.

18 PROBABILITY

A die is rolled twice. Find $P(\text{a double})$.

19 FINANCIAL MATHS

Find the simple interest on \$4000 at 3.5% p.a. for 4 years.

20 PROBLEM SOLVING

A 4 m ladder reaches 3.2 m up a wall. How far is its foot from the wall?

BONUS Show that the product of two consecutive whole numbers, plus the larger number, is always a perfect square.

WEEK 2

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & surds · Algebra · Equations · Measurement & trig · Coordinate geometry · Geometry & chance · In context · ★ Year 10 preview (Stage 5 path) · No calculator

1 NUMBER & PERCENTAGES

Write $0.\dot{4}$ (that is, $0.444\ldots$) as a fraction.

2 SCIENTIFIC NOTATION

Evaluate $(6 \times 10^5) \times (4 \times 10^{-2})$ in scientific notation.

3 INDICES

Evaluate $27^{\frac{2}{3}}$

4 SURDS

Simplify $3\sqrt{2} + \sqrt{8}$

5 ALGEBRAIC FRACTIONS

Simplify $\dfrac{4x}{9} \times \dfrac{3}{2x}$

6 EXPAND

Expand $(2x - 3)^2$

7 FACTORISE

Factorise $x^2 - 49$

8 LINEAR EQUATIONS

Solve $5 - 2(x - 4) = 3x - 7$

9 QUADRATIC EQUATIONS

Solve $x^2 = 5x$

10 FORMULAS & INEQUALITIES ★

Solve $3x - 4 > 11$

11 PYTHAGORAS

Find the exact length of the diagonal of a square with sides 6 cm.

12 TRIGONOMETRY

A 50 m kite string makes an angle of $35°$ with the ground. Use $\sin 35° \approx 0.57$ to find the height of the kite.

13 SURFACE AREA & VOLUME

Find the surface area of a closed cylinder with radius 2 cm and height 6 cm, in terms of π.

14 DISTANCE & MIDPOINT

Find the midpoint of the interval joining $(-4,\ 9)$ and $(6,\ -3)$.

15 GRADIENT

Find the gradient of the line $3x + 2y = 12$.

16 EQUATION OF A LINE

Find the equation of the line through $(1,\ 2)$ and $(3,\ 10)$.

17 GEOMETRY

Two similar triangles have matching sides 6 cm and 15 cm. A side of the smaller triangle is 4 cm. Find the matching side of the larger one.

18 PROBABILITY

In a class of 30, 18 play sport, 12 play music and 5 do both. How many do neither?

19 FINANCIAL MATHS

A surf shop buys a board for $250 and sells it for $340. Find the profit as a percentage of the cost price.

20 PROBLEM SOLVING

Two cyclists leave the same point. One rides north at 12 km/h and the other east at 16 km/h. How far apart are they after 2 hours?

BONUS Simplify $\sqrt{50} + \sqrt{18} - \sqrt{32}$

WEEK 3

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & surds · Algebra · Equations · Measurement & trig · Coordinate geometry · Geometry & chance · In context · ★ Year 10 preview (Stage 5 path) · No calculator

1 NUMBER & PERCENTAGES

A price rises by 20%, then falls by 20%. What is the overall percentage change?

2 SCIENTIFIC NOTATION

Round $0.007\,164\,9$ to 3 significant figures.

3 INDICES

Simplify $(2a^{-3})^2$, writing your answer with positive indices.

4 SURDS

Expand and simplify $\sqrt{3}(\sqrt{12}+2)$

5 ALGEBRAIC FRACTIONS ★

Simplify $\dfrac{x+1}{2}-\dfrac{x-3}{5}$

6 EXPAND

Expand $(x+5)(x-5)$

7 FACTORISE

Factorise $2x^2+7x+3$

8 LINEAR EQUATIONS

Solve $\dfrac{x}{2}-\dfrac{x}{5}=3$

9 QUADRATIC EQUATIONS

Solve $(x-3)(2x+1)=0$

10 FORMULAS & INEQUALITIES

Make r the subject of $A=\pi r^2$ (where $r>0$).

11 PYTHAGORAS

A right-angled triangle has hypotenuse 10 cm and one side 5 cm. Find the other side in simplest surd form.

12 TRIGONOMETRY ★

In a right-angled triangle, the side opposite θ is 7 cm and the hypotenuse is 14 cm. Find θ.

13 SURFACE AREA & VOLUME

A rectangular prism has volume 120 cm³, length 6 cm and width 4 cm. Find its surface area.

14 DISTANCE & MIDPOINT

Find the exact distance between $(1,\ -2)$ and $(4,\ 4)$.

15 GRADIENT

Find the gradient of a line perpendicular to the line joining $(0,\ 0)$ and $(2,\ 6)$.

16 EQUATION OF A LINE ★

Write $2x-3y+6=0$ in gradient–intercept form.

17 GEOMETRY

An exterior angle of a triangle is 125°. One of the two interior opposite angles is 60°. Find the other.

18 PROBABILITY

A bag has 5 red and 3 green balls. Two are drawn without replacement. Find $P(\text{one of each colour})$.

19 FINANCIAL MATHS ★

\$2000 is invested at 5% p.a. compound interest. Find the interest earned in 2 years.

20 PROBLEM SOLVING

A rectangular garden is 3 m longer than it is wide, and its area is 40 m². Find its dimensions.

BONUS Find two consecutive positive odd numbers whose product is 143.

WEEK 4

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & surds · Algebra · Equations · Measurement & trig · Coordinate geometry · Geometry & chance · In context · ★ Year 10 preview (Stage 5 path) · No calculator

1 NUMBER & PERCENTAGES

After a 12% discount, a bike costs \$440. Find the original price.

2 SCIENTIFIC NOTATION

Evaluate $(9 \times 10^{-3}) \div (3 \times 10^{4})$ in scientific notation.

3 INDICES

Evaluate $16^{-\frac{3}{4}}$

4 SURDS ★

Rationalise the denominator of $\dfrac{6}{\sqrt{3}}$

5 ALGEBRAIC FRACTIONS ★

Simplify $\dfrac{x^2 - 9}{x + 3}$

6 EXPAND

Expand and simplify $(x + 3)^2 - (x - 1)^2$

7 FACTORISE

Factorise $3x^2 - 27$

8 LINEAR EQUATIONS

Solve $\dfrac{2x - 1}{3} - \dfrac{x + 2}{4} = 1$

9 QUADRATIC EQUATIONS ★

Solve $x^2 + 6x + 4 = 0$ by completing the square.

10 FORMULAS & INEQUALITIES ★

Solve $-2x + 5 \leq 11$

11 PYTHAGORAS ★

A box is 3 cm by 4 cm by 12 cm. Find the length of the longest rod that fits inside it.

12 TRIGONOMETRY

A ship sails 8 km on a bearing of $180°$T, then 6 km on a bearing of $270°$T. How far is it from its start?

13 SURFACE AREA & VOLUME ★

Find the volume of a sphere with radius 3 cm, in terms of π.

14 DISTANCE & MIDPOINT

The midpoint of AB is $(2,\ -1)$ and A is $(-3,\ 4)$. Find B.

15 GRADIENT

The line through $(k,\ 3)$ and $(5,\ 9)$ has gradient 2. Find k.

16 EQUATION OF A LINE

Find the equation of the line through $(2,\ -1)$ that is parallel to $y = 3x + 5$.

17 GEOMETRY ★

An angle at the centre of a circle is $110°$. Find the angle at the circumference standing on the same arc.

18 PROBABILITY ★

Of 50 students, 30 play netball, and 12 of the netballers also play touch. A netballer is chosen at random. Find $P(\text{plays touch})$.

19 FINANCIAL MATHS

Kim earns \$32 an hour with time-and-a-half for overtime. Find her pay for 38 normal hours and 6 overtime hours.

20 PROBLEM SOLVING

The sum of a number and its reciprocal is $\frac{10}{3}$. Find the number.

BONUS Simplify $\dfrac{1}{\sqrt{3} - \sqrt{2}}$

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & surds · Algebra · Equations · Measurement & trig · Coordinate geometry · Geometry & chance · In context · ★ Year 10 preview (Stage 5 path) · No calculator

1 NUMBER & PERCENTAGES

Convert 72 km/h to metres per second.

2 SCIENTIFIC NOTATION

A dust particle has mass 7.5×10^{-10} kg. How many particles have a total mass of 1.5×10^{-6} kg?

3 INDICES

Simplify $\dfrac{x^3y^{-2}}{x^{-1}y^4}$, with positive indices.

4 SURDS

Expand and simplify $(\sqrt{5}+2)(\sqrt{5}-2)$

5 ALGEBRAIC FRACTIONS

Simplify $\dfrac{3}{x}+\dfrac{2}{3x}$

6 EXPAND ★

Expand and simplify $(x+2)(x^2-x+3)$

7 FACTORISE

Factorise $ax+ay+3x+3y$

8 LINEAR EQUATIONS

Solve $4(3-x)=2(x+9)$

9 QUADRATIC EQUATIONS ★

Use the quadratic formula to solve $x^2-2x-4=0$.

10 FORMULAS & INEQUALITIES

Make u the subject of $v^2=u^2+2as$.

11 PYTHAGORAS

Rectangle $ABCD$ has $AB=8$ cm and diagonal $AC=10$ cm. Find its area.

12 TRIGONOMETRY ★

In $\triangle ABC$, $a=10$, $A=30°$ and $B=45°$. Use the sine rule to find b exactly.

13 SURFACE AREA & VOLUME ★

A cone has radius 6 cm and slant height 10 cm. Find its perpendicular height and its volume, in terms of π.

14 DISTANCE & MIDPOINT

Find the side lengths of the triangle with vertices $A(0,\,0)$, $B(4,\,0)$ and $C(2,\,2\sqrt{3})$. What type of triangle is it?

15 GRADIENT ★

Find the angle a line with gradient 1 makes with the positive x-axis.

16 EQUATION OF A LINE ★

Find the equation of the perpendicular bisector of the interval joining $(1,\,1)$ and $(5,\,3)$.

17 GEOMETRY

Each exterior angle of a regular polygon is $24°$. How many sides does it have?

18 PROBABILITY

Three coins are tossed. Find $P(\text{exactly two heads})$.

19 FINANCIAL MATHS ★

A \$50 000 machine depreciates by 10% p.a. Find its value after 2 years.

20 PROBLEM SOLVING

Lily walks to the beach at 5 km/h and jogs home at 10 km/h. The round trip takes 1.5 hours. How far away is the beach?

BONUS Show that $x^2+6x+10$ is positive for every value of x.

WEEK 6

Summer Maths

Ready for Year 10

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & surds · Algebra · Equations · Measurement & trig · Coordinate geometry · Geometry & chance · In context · ★ Year 10 preview (Stage 5 path) · No calculator

1 NUMBER & PERCENTAGES

Write $0.\dot{1}\dot{2}$ (that is, $0.121212\ldots$) as a fraction.

2 SCIENTIFIC NOTATION

Light travels at 3×10^8 m/s. How far does it travel in 5×10^2 seconds?

3 INDICES ★

Evaluate $\log_2 32 + \log_3 \frac{1}{9}$

4 SURDS

Evaluate $(2\sqrt{3})^2 + \sqrt{27} \times \sqrt{3}$

5 ALGEBRAIC FRACTIONS ★

Simplify $\dfrac{x^2 + 5x + 6}{x^2 - 4}$

6 EXPAND

Expand and simplify $(2x + 3)(2x - 3) - (2x - 1)^2$

7 FACTORISE

Factorise $6x^2 - x - 2$

8 LINEAR EQUATIONS ★

Solve simultaneously: $3x + 2y = 16$ and $x - y = 2$

9 QUADRATIC EQUATIONS

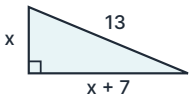
Solve $2x^2 - 5x - 3 = 0$

10 FORMULAS & INEQUALITIES ★

Solve $5 - 3x < 2x - 10$

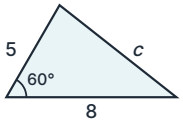
11 PYTHAGORAS

Find x. (cm)



12 TRIGONOMETRY ★

Find c using the cosine rule.



13 SURFACE AREA & VOLUME ★

A hemisphere of radius 3 cm sits on top of a cylinder with radius 3 cm and height 10 cm. Find the total volume, in terms of π.

14 DISTANCE & MIDPOINT

Find the equation of the circle with centre $(0,\ 0)$ that passes through $(3,\ 4)$.

15 GRADIENT

The points $(1,\ 2)$, $(3,\ 8)$ and $(k,\ 14)$ lie on a straight line. Find k.

16 EQUATION OF A LINE ★

Find the equation, in general form, of the line through $(-1,\ 4)$ with gradient $-\frac{2}{3}$.

17 GEOMETRY ★

OT is a radius and PT is a tangent to the circle at T. If $\angle TOP = 50°$, find $\angle OPT$.

18 PROBABILITY

A spinner lands on red with probability 0.3. It is spun twice. Find $P(\text{at least one red})$.

19 FINANCIAL MATHS

Ben earns \$1100 a week. He takes 4 weeks' annual leave with 17.5% leave loading. Find his total holiday pay.

20 PROBLEM SOLVING

A farmer uses 40 m of fencing for three sides of a rectangular pen against a wall. The area is 200 m². Find the dimensions.

BONUS Find all real solutions of $x^4 - 13x^2 + 36 = 0$.

Hints & formulas

Number

Increase by 20% then decrease by 20%: $\times 1.2 \times 0.8$

Original price = sale price ÷ (decimal you paid)

Recurring decimals: $0.\dot{4} = \frac{4}{9}$, $0.\dot{1}\dot{2} = \frac{12}{99}$

Scientific notation: $a \times 10^n$, $1 \leq a < 10$

Indices & surds

$a^{-n} = \frac{1}{a^n} \cdot a^{\frac{m}{n}} = (\sqrt[n]{a})^m$

$\sqrt{ab} = \sqrt{a}\sqrt{b}$, e.g. $\sqrt{48} = 4\sqrt{3}$

$(\sqrt{a}+b)(\sqrt{a}-b) = a - b^2$

★ $\frac{6}{\sqrt{3}} = \frac{6\sqrt{3}}{3} \cdot \log_a x = n \Leftrightarrow a^n = x$

Algebra

Fractions: common denominator, brackets around numerators

$(a \pm b)^2 = a^2 \pm 2ab + b^2 \cdot (a+b)(a-b) = a^2 - b^2$

Factorise: HCF first, then DOTS, trinomials or grouping

Cancel factors, never terms

Equations

Fractions: multiply every term by the common denominator

$x^2 = 5x$: move everything to one side and factorise

★ $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ · complete the square: add $(\frac{b}{2})^2$

★ Inequalities: ×/÷ by a negative reverses the sign

Measurement & trig

$\sin\theta = \frac{O}{H} \cdot \cos\theta = \frac{A}{H} \cdot \tan\theta = \frac{O}{A}$

$\sin 30° = \cos 60° = \frac{1}{2} \cdot \sin 45° = \frac{\sqrt{2}}{2}$

Cylinder $SA = 2\pi r^2 + 2\pi rh$

★ $\frac{a}{\sin A} = \frac{b}{\sin B} \cdot c^2 = a^2 + b^2 - 2ab\cos C$

★ Cone $V = \frac{1}{3}\pi r^2 h$ · sphere $V = \frac{4}{3}\pi r^3$

Coordinate geometry

$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2} \cdot M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$m = \frac{y_2-y_1}{x_2-x_1}$ · parallel: equal m · perpendicular: $m_1 m_2 = -1$

★ $y - y_1 = m(x - x_1)$ · general form $ax + by + c = 0$

★ $m = \tan\theta$ (angle with the x-axis)

Geometry & chance

Polygon angle sum $(n-2) \times 180°$ · exterior angles add to $360°$

Similar figures: multiply by the scale factor

Venn: $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

★ Angle at the centre is twice the angle at the circumference; tangent ⊥ radius

In context

$I = Prn$ · profit % = $\frac{\text{profit}}{\text{cost}} \times 100\%$

Leave loading: 17.5% extra on holiday pay

★ $A = P(1+r)^n$ · depreciation $S = V_0(1-r)^n$

Worded problems: define the unknown, write an equation, check the answer makes sense

Answers

1 Week 1 answers

1 **$306** 360×0.85

2 5.8×10^{-5} move the decimal point 5 places right

3 $27x^6y^3$ cube each factor

4 $4\sqrt{3}$ $\sqrt{16 \times 3}$

5 $\frac{7x}{12}$ $\frac{4x}{12} + \frac{3x}{12}$

6 $x^2 + 4x - 21$ $x^2 - 3x + 7x - 21$

7 $(x + 4)(x + 5)$ $4 \times 5 = 20$, $4 + 5 = 9$

8 $x = 5$ $x + 3 = 4x - 12$, $3x = 15$

9 $x = 2$ **or** $x = 5$ $(x - 2)(x - 5) = 0$

10 $h = \dfrac{V}{\pi r^2}$ divide both sides by πr^2

11 $x = \sqrt{13}$ $x^2 = 4 + 9$

12 $x = 9$ **cm** $\cos 60° = \frac{x}{18}$

13 100π **cm³** $\pi \times 25 \times 4$

14 **10 units** $\sqrt{6^2 + 8^2}$

15 -2 $\frac{-5-7}{4-(-2)} = \frac{-12}{6}$

16 $y = 2x + 3$ $5 = 2(1) + c$, $c = 3$

17 $120°$ angle sum $(6 - 2) \times 180 = 720$; $720 \div 6$

18 $\frac{1}{6}$ 6 doubles out of 36

19 **$560** $I = 4000 \times 0.035 \times 4$

20 **2.4 m** $\sqrt{16 - 10.24} = \sqrt{5.76}$

B $n(n + 1) + (n + 1) = (n + 1)^2$ take out the common factor $(n + 1)$

2 Week 2 answers

1 $\frac{4}{9}$ $10x - x = 4$, so $9x = 4$

2 2.4×10^4 24×10^3

3 9 $(\sqrt[3]{27})^2 = 3^2$

4 $5\sqrt{2}$ $\sqrt{8} = 2\sqrt{2}$

5 $\frac{2}{3}$ $\frac{12x}{18x}$

6 $4x^2 - 12x + 9$ $(2x)^2 - 2(2x)(3) + 9$

7 $(x - 7)(x + 7)$ difference of two squares

8 $x = 4$ $13 - 2x = 3x - 7$, $20 = 5x$

9 $x = 0$ **or** $x = 5$ $x(x - 5) = 0$ – don't divide by x

10 $x > 5$ $3x > 15$

11 $6\sqrt{2}$ **cm** $\sqrt{36 + 36} = \sqrt{72}$

12 ≈ 28.5 **m** $h = 50 \times 0.57$

13 32π **cm²** $2\pi(4) + 2\pi(2)(6) = 8\pi + 24\pi$

14 $(1,\ 3)$ $(\frac{-4+6}{2},\ \frac{9-3}{2})$

15 $-\frac{3}{2}$ $y = -\frac{3}{2}x + 6$

16 $y = 4x - 2$ $m = \frac{8}{2} = 4$; $2 = 4 + c$

17 **10 cm** scale factor $\frac{15}{6} = 2.5$; 4×2.5

18 **5** $30 - (18 + 12 - 5)$

19 36% $\frac{90}{250} \times 100\%$

20 **40 km** 24 km and 32 km: $\sqrt{576 + 1024}$

B $4\sqrt{2}$ $5\sqrt{2} + 3\sqrt{2} - 4\sqrt{2}$

Answers

3 Week 3 answers

1 **a 4% decrease** $1.2 \times 0.8 = 0.96$

2 $0.007\,16$ figures 7, 1, 6; the next digit is 4

3 $\frac{4}{a^6}$ $4a^{-6}$

4 $6 + 2\sqrt{3}$ $\sqrt{36} + 2\sqrt{3}$

5 $\frac{3x+11}{10}$ $\frac{5(x+1)-2(x-3)}{10}$

6 $x^2 - 25$ difference of two squares

7 $(2x + 1)(x + 3)$ check: $2x^2 + 6x + x + 3$

8 $x = 10$ $\frac{3x}{10} = 3$

9 $x = 3$ **or** $x = -\frac{1}{2}$ one of the factors must be 0

10 $r = \sqrt{\frac{A}{\pi}}$ $r^2 = \frac{A}{\pi}$

11 $5\sqrt{3}$ **cm** $\sqrt{100 - 25} = \sqrt{75}$

12 $\theta = 30°$ $\sin\theta = \frac{1}{2}$

13 **148 cm²** height 5; $2(24 + 30 + 20)$

14 $3\sqrt{5}$ $\sqrt{9 + 36} = \sqrt{45}$

15 $-\frac{1}{3}$ that line has gradient 3

16 $y = \frac{2}{3}x + 2$ $3y = 2x + 6$

17 $65°$ exterior angle = sum of interior opposite angles

18 $\frac{15}{28}$ $\frac{5}{8} \times \frac{3}{7} + \frac{3}{8} \times \frac{5}{7}$

19 **$205** $2000 \times 1.05^2 = 2205$

20 **5 m by 8 m** $x(x + 3) = 40$, $(x + 8)(x - 5) = 0$

B **11 and 13** $n(n + 2) = 143$

4 Week 4 answers

1 **$500** $440 \div 0.88$

2 3×10^{-7} $9 \div 3$; 10^{-3-4}

3 $\frac{1}{8}$ $\frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3}$

4 $2\sqrt{3}$ $\frac{6\sqrt{3}}{3}$

5 $x - 3$ $\frac{(x-3)(x+3)}{x+3}$

6 $8x + 8$ $(x^2 + 6x + 9) - (x^2 - 2x + 1)$

7 $3(x - 3)(x + 3)$ HCF 3, then difference of squares

8 $x = \frac{22}{5}$ $4(2x - 1) - 3(x + 2) = 12$, $5x = 22$

9 $x = -3 \pm \sqrt{5}$ $(x + 3)^2 = 5$

10 $x \geq -3$ $-2x \leq 6$; dividing by -2 reverses the sign

11 **13 cm** $\sqrt{9 + 16 + 144} = \sqrt{169}$

12 **10 km** 8 km south then 6 km west: $\sqrt{64 + 36}$

13 36π **cm³** $\frac{4}{3}\pi \times 27$

14 $B(7,\ -6)$ x: $-3 \to 2 \to 7$; y: $4 \to -1 \to -6$

15 $k = 2$ $\frac{6}{5-k} = 2$

16 $y = 3x - 7$ $-1 = 3(2) + c$

17 $55°$ half the angle at the centre

18 $\frac{2}{5}$ $\frac{12}{30}$

19 **$1504** $38 \times 32 + 6 \times 48 = 1216 + 288$

20 **3 or** $\frac{1}{3}$ $3x^2 - 10x + 3 = 0$, $(3x - 1)(x - 3) = 0$

B $\sqrt{3} + \sqrt{2}$ multiply top and bottom by $\sqrt{3} + \sqrt{2}$; the bottom becomes $3 - 2 = 1$

Answers

5 Week 5 answers

1 **20 m/s** $72\,000\text{ m} \div 3600\text{ s}$

2 2×10^3 $\frac{1.5}{7.5} \times 10^4 = 0.2 \times 10^4$

3 $\frac{x^4}{y^6}$ $x^{3-(-1)}y^{-2-4}$

4 1 $5 - 4$

5 $\frac{11}{3x}$ $\frac{9}{3x} + \frac{2}{3x}$

6 $x^3 + x^2 + x + 6$ $x^3 - x^2 + 3x + 2x^2 - 2x + 6$

7 $(x + y)(a + 3)$ $a(x + y) + 3(x + y)$

8 $x = -1$ $12 - 4x = 2x + 18$, $-6 = 6x$

9 $x = 1 \pm \sqrt{5}$ $\frac{2 \pm \sqrt{4+16}}{2} = \frac{2 \pm 2\sqrt{5}}{2}$

10 $u = \pm\sqrt{v^2 - 2as}$ $u^2 = v^2 - 2as$

11 **48 cm²** $BC = \sqrt{100 - 64} = 6$; 8×6

12 $b = 10\sqrt{2}$ $b = \frac{10 \sin 45^\circ}{\sin 30^\circ} = \frac{10 \times \frac{\sqrt{2}}{2}}{\frac{1}{2}}$

13 **8 cm; 96π cm³** $h = \sqrt{100 - 36}$; $V = \frac{1}{3}\pi(36)(8)$

14 **all 4 units: equilateral** $AC = \sqrt{4 + 12} = 4$, $BC = \sqrt{4 + 12} = 4$

15 45° $\tan\theta = m = 1$

16 $y = -2x + 8$ midpoint $(3, 2)$; gradient $\frac{1}{2}$, so use -2

17 **15** $360 \div 24$

18 $\frac{3}{8}$ HHT, HTH, THH out of 8

19 **$40 500** $50\,000 \times 0.9^2$

20 **5 km** $\frac{d}{5} + \frac{d}{10} = 1.5$

B $x^2 + 6x + 10 = (x + 3)^2 + 1 \geq 1$ a square is never negative

6 Week 6 answers

1 $\frac{4}{33}$ $100x - x = 12$, $x = \frac{12}{99}$

2 1.5×10^{11} **m** 15×10^{10}

3 3 $5 + (-2)$

4 21 $12 + \sqrt{81}$

5 $\frac{x+3}{x-2}$ $\frac{(x+2)(x+3)}{(x-2)(x+2)}$

6 $4x - 10$ $4x^2 - 9 - (4x^2 - 4x + 1)$

7 $(2x + 1)(3x - 2)$ check: $6x^2 - 4x + 3x - 2$

8 $x = 4, y = 2$ $x = y + 2$: $3y + 6 + 2y = 16$

9 $x = 3$ **or** $x = -\frac{1}{2}$ $(2x + 1)(x - 3) = 0$

10 $x > 3$ $15 < 5x$

11 $x = 5$ $x^2 + (x + 7)^2 = 169$, so $x^2 + 7x - 60 = 0$

12 $c = 7$ $c^2 = 25 + 64 - 2(5)(8)\cos 60^\circ = 49$

13 108π **cm³** $\frac{2}{3}\pi(27) + \pi(9)(10) = 18\pi + 90\pi$

14 $x^2 + y^2 = 25$ radius $= \sqrt{9 + 16} = 5$

15 $k = 5$ gradient 3: $\frac{14-2}{k-1} = 3$

16 $2x + 3y - 10 = 0$ $y - 4 = -\frac{2}{3}(x + 1)$, so $3y - 12 = -2x - 2$

17 40° the tangent is perpendicular to the radius: $180 - 90 - 50$

18 0.51 $1 - 0.7^2 = 1 - 0.49$

19 **$5170** 4400×1.175

20 **10 m by 20 m** $x(40 - 2x) = 200$, $(x - 10)^2 = 0$

B $x = \pm 2$ **or** $x = \pm 3$ $(x^2 - 4)(x^2 - 9) = 0$